

NEHA DABEERU

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PROFESSIONAL SUMMARY

Python Backend Software Engineer with 4+ years of experience building production APIs, event-driven services, and distributed data workflows on AWS. Experienced in designing GraphQL and REST APIs, orchestrating asynchronous processing with Lambda, Step Functions, EventBridge, and SQS, and developing reliable persistence layers with PostgreSQL and DynamoDB. Proven track record of resolving production failures, implementing idempotent and retry-safe workflows, and improving system reliability through automated testing, observability, and failure recovery.

PROFESSIONAL EXPERIENCE

Software Engineer, Data & AI Platforms, Merck Sharp & Dohme (New Jersey, USA - REMOTE)

Jan 2025 – Present

- Built AppSync GraphQL APIs with Python Lambda & VTL resolvers for **2K+** users across encoding, approval, audit, and external-partner workflows.
- Led development of a specification platform, replacing manual Excel handoffs with versioned approvals, RBAC, audit trails & structured states.
- Resolved AppSync timeouts from **30+** sec Databricks cold starts via async workflow DynamoDB Streams, Step Functions, & GraphQL subscriptions.
- Created hybrid BM25 & neural k-NN auto-coding with score normalization, weighted ranking & tie-breaking, reaching **99.52%** top-match accuracy.
- Automated event-driven ingestion with Step Functions, Databricks, S3 & OpenSearch, enabling **5–6** hour incremental syncs and nightly full loads.
- Designed append-only DynamoDB audit logs with Global Tables replication, tracking user actions and state changes across regulated workflows.
- Streamlined MuleSoft & GraphQL subscription integrations for real-time clinical updates, reducing manual corrections and status tracking.
- Created 12-scenario E2E regression suite using pytest, moto, and live AWS integration tests, preventing recurrence of prod data-integrity defect.

Software Engineer II (CONTRACT), Bank of America (North Carolina, USA)

Jun 2024 – Aug 2024

- Built checksum reconciliation framework to detect schema drift, missing records, and silent data mismatches for Cards and Payments migration.
- Automated source-to-target validation for referential integrity, data types, record counts, and field values, reducing pre-cutover verification effort.
- Generated exception reports covering mismatches, affected records & failure summaries, speeding defect triage, remediation & cutover decisions.

Software Engineer I, Bank of America (Tamil Nadu, India)

Jun 2020 – Jan 2023

- Designed distributed processing framework for financial-risk workloads, partitioning execution across sharded datasets reducing runtime by **60%**.
- Eliminated idle compute and wasted capacity by replacing always-on processing with event-triggered workflows, cutting infrastructure costs **35%**.
- Reduced duplicate integration logic through reusable cross-database components for connection management, query execution & error handling.
- Cut manual migration effort **70–80%** with a rule-based SQL translation service that parsed legacy syntax and generated target-dialect queries.
- Improved workflow reliability with idempotent processing, checkpoint-based recovery, & restart-safe execution, resuming failed jobs latest stage.

Software Engineer Intern, Rashtriya Ispat Nigam Limited (Andhra Pradesh, India)

Jul 2017 – Sep 2017

- Developed software using advanced Python concepts, optimizing automated trade rebalancing using time-series data for **1K+** transactions/day.
- Built Django-React validation framework with schema-driven checks & client-side rules engine, enhancing MiFID regulatory tax compliance.

EDUCATION

M.S., Data Analytics, San Jose State University, California

Jan 2023 – Dec 2024

Relevant Coursework: Cloud Computing, Database Systems, Data Engineering, Machine Learning, Big Data Analytics, Statistical Modeling

B.E., Computer Science and Engineering, SRM University, India

June 2016 – May 2020

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Distributed Systems, Web Technologies

SELECTED ENGINEERING PROJECTS

DataFlo & DQ Checker — GenAI Clinical Data Platform:

- Cut ETL onboarding from 8–10 weeks to 1 week with GenAI auto-mapping, semantic retrieval, prompt-based reranking & data standardization.
- Developed multi-model RAG across Bedrock Haiku & OpenAI with confidence scoring and fallback chains, reaching **90%+** auto-mapping accuracy.
- Eliminated manual authoring for **100+** data-quality checks with LLM natural-language rule engine with validation, real-time alerts, & audit trails.

MedPredict — Hospital Readmission Prediction Platform:

- Reached **92%+** readmission accuracy with transformer embeddings, FAISS retrieval, and PyTorch classification across 40K+ MIMIC-III records.
- Cut inference latency to 2 secs by deploying containerized FastAPI service on AWS with async request handling and model-serving optimizations.

TECHNICAL SKILLS

Languages and Backend: Python, Java, SQL, JavaScript, TypeScript, REST APIs, GraphQL, FastAPI, AWS AppSync, API Gateway, AWS Lambda, VTL Resolvers, API Design, Authentication, Authorization

Cloud, Distributed Systems, and Databases: AWS, Step Functions, EventBridge, SQS, Event-Driven Architecture, Asynchronous Processing, Workflow Orchestration, Idempotency, Retry Strategies, Failure Recovery, Concurrency Control, PostgreSQL, MySQL, DynamoDB, Amazon OpenSearch, pgvector, Delta Lake

DevOps, Testing, Observability, and AI: pytest, moto, Docker, Terraform, GitHub Actions, CI/CD, Git, CloudWatch, X-Ray, Structured Logging, Distributed Tracing, AWS Bedrock, RAG, Vector Search, Apache Spark, PySpark, Databricks

CERTIFICATION

AWS Certified Data Engineer – Associate, Amazon Web Services

May 2025 – May 2028